

Inter-Scale evidence of an Oriented Angular Transport in Semiprime Geometry

Abstract

We report a multi-scale geometric analysis of an oriented angular transport phenomenon arising in the factorization geometry of semiprimes $n = pq$. Using a pendulum-like construction based on the chords $n \rightarrow p$ and $p \rightarrow q$, we define three angular variables θ_n, α, β measured in a shared geometric reference frame.

Across increasing dyadic shells (KN64, KN256, KN1024), we observe a persistent asymmetric splitting of the conditional distribution of β , governed by the geometric position of p relative to the tangent frame induced by n .

A purely geometric partition based on the sign of $\cos(\alpha)$ reveals a stable, scale-invariant angular transport mechanism, incompatible with binning artefacts or finite-sample effects.

1. Geometric Framework and Definitions

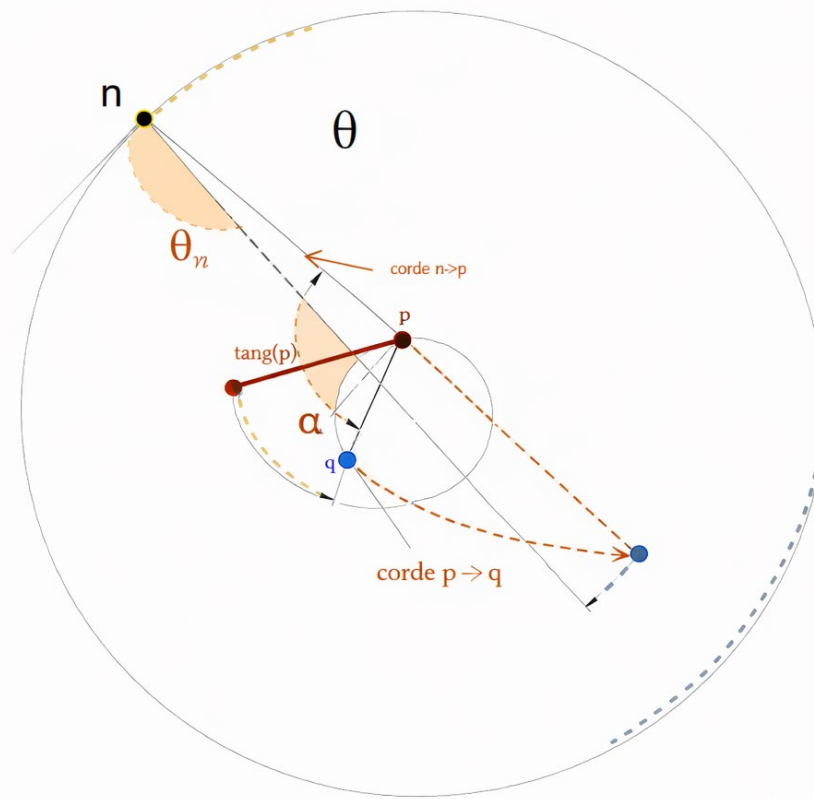


Figure 1 illustrates the geometric construction and angular definitions used throughout this work.

Let $n = p \cdot q$ be a semiprime with $p \leq q$.

The geometric system is defined as follows:

- The point n defines the origin of the pendulum system.
- The chord $n \rightarrow p$ defines the primary geometric axis.

- The **perpendicular to the chord $n \rightarrow p$ at point n** defines the common angular reference frame.
- θ_n is the angle between the radius of n and this perpendicular.
- α is the angle between this perpendicular and the tangent at point p .
- β is the angle between the same perpendicular and the tangent at point q .

Crucially, α and β are measured in the **same reference frame**, making conditional angular analysis geometrical meaningful.

2. Geometric Partition: Lobes and Mid-Band

To avoid any arbitrary angular threshold, the partition is defined **purely geometrically**:

- **Positive lobe (pos-lobe)**

$$\cos(\alpha) \geq 0$$

- **Negative lobe (neg-lobe)**

$$\cos(\alpha) < 0$$

This partition :

- corresponds to a division of the plane into two half-planes,
- is invariant under angular wrapping ($\pm\pi$),
- is independent of histogram binning, resolution, or sample size.

An optional **mid-band** is defined by:

$$|\cos(\alpha)| < \varepsilon$$

Geometrically, this corresponds to near-orthogonality with respect to the reference axis and represents a pendulum transition zone, typically sparsely populated.

3. Empirical Results

3.1 Distribution of θ_n

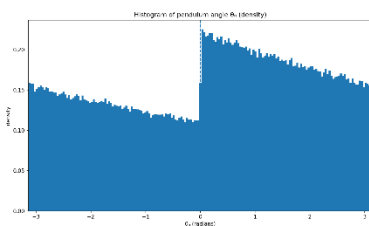


Figure 2a θ_n on 2^{64}

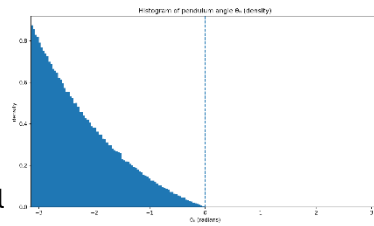


Figure 2b θ_n on 2^{256}

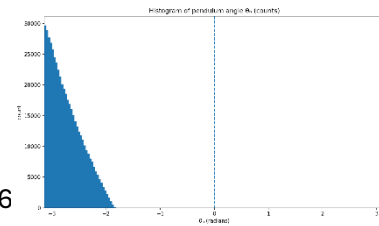


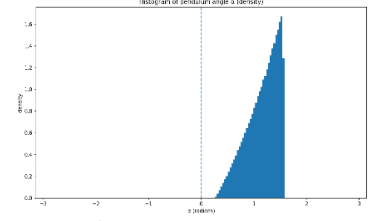
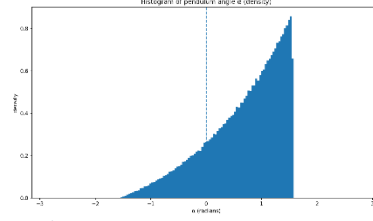
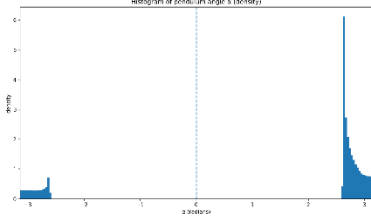
Figure 2c θ_n on 2^{1024}

Across all scales considered (KN64, KN256, KN1024), the distribution of θ_n is :

- strongly non-uniform,
- confined to a limited angular support,
- stable in shape under scale increase.

This indicates that n is not angularly isotropic in this reference frame.

3.2 Distribution of α



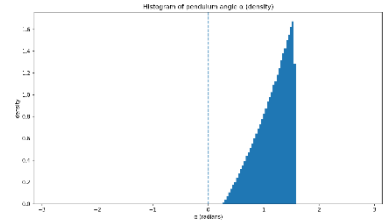
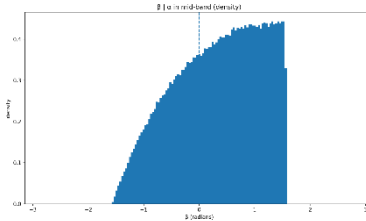
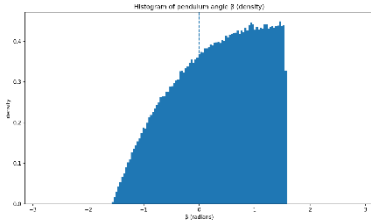
The angle α exhibits :

- strict confinement to a single half-plane,
- increasing sharpness as the dyadic scale increases,
- a marked depletion near the mid-band.

This behavior retrospectively validates the partition based on $\cos(\alpha)$.

This confinement retrospectively validates the cosine-based lobe partition, which emerges naturally from the geometry.

3.3 Marginal Distribution of β



The marginal distribution of β is asymmetric but does not, by itself, exhibit a clear bifurcation.

This is a key observation: **the phenomenon is not visible without conditioning.**

The absence of clear bifurcation in the marginal distribution indicates that the phenomenon is not intrinsic to β alone but arises only through geometric conditioning.

3.4 Conditional Distribution of β Given the Lobe of α

Before introducing the conditional analysis, an important scale-dependent change in the shape of the β -distribution must be highlighted.

While the marginal and mid-band–restricted distributions of β remain broadly comparable between KN64 and KN256, a clear qualitative change is observed at KN1024. In particular, the curvature of the β -density in the mid-band transitions from a positively rounded profile at lower scales to a negatively rounded profile at the highest scale.

This inversion of concavity is not accompanied by a comparable change in the distributions of θ_n or α , nor does it arise uniformly across all configurations. Instead, it anticipates a regime change that becomes explicit only after conditioning on the geometric lobe defined by the sign of $\cos(\alpha)$.

This observation suggests that the angular transport mechanism undergoes a scale-dependent transition, motivating a closer examination of β under geometric conditioning.

When conditioning β on the sign of $\cos(\alpha)$, we observe:

- a **clear separation of the distributions** between pos-lobe and neg-lobe,
- a persistent asymmetry,
- progressive sharpening of the lobes as the dyadic scale increases,
- remarkable stability of angular supports.

From KN64 to KN1024, the phenomenon:

- does not vanish,
- does not symmetrize,
- instead **refines and stabilizes**.

When conditioning the distribution of β on the geometric lobe defined by $\cos(\alpha)$, a clear asymmetric bifurcation emerges...

4. Interpretation: Inter-Scale Pendulum Transport

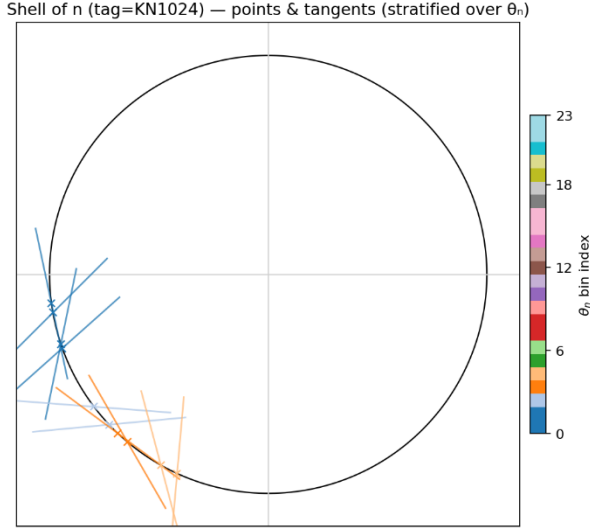
The observations converge toward a single interpretation:

- p acts as a **geometric articulation point**,
- the forward/backward position of p determines distinct angular regimes for q ,
- angular information is **transported** from n to q through p ,
- this transport is **oriented, non-random, and scale-invariant**.

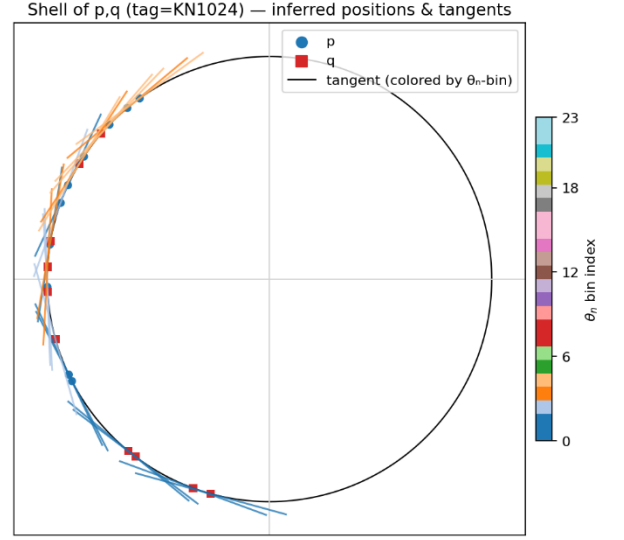
A statistical artefact would be expected to diminish or wash out as the scale increases. Here, the opposite behavior is observed.

The system behaves as an oriented angular transport mechanism, in which the geometric position of p acts as an articulation point between the angular states of n and q .

5. Conclusion



Each point represents a semiprime n positioned on the unit circle according to its angular coordinate θ_n . The associated tangent at each point corresponds to the local direction induced by the chord $n \rightarrow p$. Samples are stratified uniformly over the angular range of θ_n to ensure balanced coverage of the shell. Tangent segments are color-coded by the θ_n -bin index, indicating the angular provenance of each configuration.



For each sampled semiprime n , the positions of its prime factors p (circles) and q (squares) are inferred on the unit circle from their respective tangent angles. Tangent segments represent the local directions at p and q , expressed in the same angular reference frame as the source shell. Colors match those used in the shell of n , preserving the one-to-one correspondence between source and transported configurations. This figure visualizes the angular transport from n to its factors across the dyadic scale.

Sampling and construction of the final shell figures

The final shell figures for n and for the pair (p, q) are not exhaustive plots of the full datasets, but structured visual summaries designed to preserve geometric readability while remaining representative of the underlying distributions.

For each dyadic scale KN , a large dataset of approximately 500,000 balanced semiprimes was first generated and enriched with pendulum angles θ_n , α , and β . From this dataset, a **stratified subsample** was then extracted for visualization.

Stratified sampling over θ_n

The angular range $(-\pi, \pi]$ of θ_n was divided into a fixed number of bins (typically 24).

Within each bin, a small and equal number of semiprimes was randomly selected. This ensures that :

- all regions of the n -shell are represented,
- no angular sector dominates the visualization,
- the apparent structure is not an artefact of density variations.

The color assigned to each tangent segment corresponds to the index of the θ_n -bin from which the sample was drawn, allowing the reader to visually track the angular provenance of each configuration.

Construction of shell representations

- **Shell of n :**
Each sampled semiprime is plotted as a point on the unit circle at angular position θ_n , together with its associated tangent direction. This figure represents the “source shell” of the pendulum transport.
- **Shell of p, q :**
For the same sampled semiprimes, the positions of p and q are inferred from their tangent angles and plotted on a common shell.
Distinct markers are used to distinguish p (circles) from q (squares), while tangents are drawn using the same color coding as in the n -shell.

Importantly, **each configuration appearing in the (p, q) -shell corresponds exactly to a configuration shown in the n -shell**, preserving a one-to-one correspondence between source and transported geometry.

Purpose of the visualization

These figures are not intended to estimate densities or probabilities. Their purpose is to:

- reveal the geometric coherence of the angular transport,
- visualize the alignment and dispersion of tangents across shells,
- illustrate how angular structure propagates from n to its factors across dyadic scales.

Results

This multi-scale analysis confirms the existence of an **angular pendulum transport** in semiprime geometry. A geometric partition based on $\cos(\alpha)$ reveals an asymmetric doubling of the distribution of β that persists and sharpens with increasing dyadic scale.

The result is:

- independent of binning choices,
- invariant under angular wrapping,
- grounded in a shared reference frame,
- and constitutes a **robust geometric fact**.

This work establishes the existence of a robust, scale-invariant angular transport structure in semiprime geometry.

The results suggest that factorization induces non-trivial directional constraints, opening the way to a more formal geometric description of angular information flow in arithmetic spaces.
